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Convexity of the Le Roy parameter threshold

Research proof and Lean formalization

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Abstract

We prove the convexity question posed at the end of Section 3 of Simon's paper on a Mittag–Leffler function of Le Roy type. Rescaling the auxiliary variable expresses the literal threshold as a supremum of positive perspectives of a convex function. We prove finiteness and positivity and identify this supremum with the source's infimum over admissible positive parameters. The distinct complete-monotonicity conjecture remains open here.

Fix $\gamma > 1$. Simon [1, p. 5, Section 3] defines, for $0 < \alpha < 1/\gamma$,

$$B_\gamma(\alpha) = \inf\{\beta > 0 : z + \gamma(z^{\beta-\alpha} - z^\beta) \le 1 \text{ for every } 0 < z < 1\}.$$

He states that $\alpha \mapsto B_\gamma(\alpha)/\alpha$ is nondecreasing, and asks whether $\alpha \mapsto B_\gamma(\alpha)$ is convex. We prove the latter assertion.

Theorem 1. *For every real $\gamma > 1$, the literal function B_γ is convex on $(0, 1/\gamma)$.*

Put $f(t) = -\log(1 - e^{-t})$ for $t > 0$. Its derivative

$$f'(t) = -\frac{1}{e^t - 1}$$

is strictly increasing, so f is strictly convex. For $s > 0$ define

$$J_s(\alpha) = \alpha + \frac{\alpha}{s}\big(\log\gamma - f(s) + f(s/\alpha)\big), \qquad \alpha > 0. \tag{1}$$

Lemma 1. *For $\gamma > 1$ and $0 < \alpha < 1/\gamma$,*

$$B_\gamma(\alpha) = \sup_{s>0} J_s(\alpha), \qquad 0 < \alpha < B_\gamma(\alpha) \le \alpha + 1.$$

Proof. Set $z = e^{-t}$, $t > 0$. Taking logarithms of positive factors shows that the criterion is equivalent to

$$\beta \ge \alpha + \frac{\log\gamma + \log(1 - e^{-\alpha t}) - \log(1 - e^{-t})}{t} \quad (t > 0).$$

Substitute $s = \alpha t$ to obtain exactly $\beta \ge J_s(\alpha)$ for every $s > 0$.

For completeness, $\beta = \alpha + 1$ is admissible whenever $\gamma\alpha \le 1$: the elementary exponential inequalities give

$$\gamma e^{-t}(1 - e^{-\alpha t}) \le \gamma\alpha t e^{-t} \le t e^{-t} \le 1 - e^{-t}.$$

Thus the supremum is finite and at most $\alpha + 1$. As $s \to \infty$, $f(s) \to 0$. Choose $s > 0$ with $f(s) < \log\gamma$; since $f(s/\alpha) > 0$, equation (1) gives $J_s(\alpha) > \alpha > 0$. If $M = \sup_{s>0} J_s(\alpha)$, the set in the source's infimum is therefore exactly $[M, \infty)$. This proves the claimed equality and bounds. □

Proof of Theorem 1. For fixed $s > 0$, the function $\alpha \mapsto \alpha f(s/\alpha)$ is the classical positive perspective of f, and is convex on $(0, \infty)$. Explicitly, for $a, b > 0$, $\lambda, \mu \ge 0$, $\lambda + \mu = 1$, and $c = \lambda a + \mu b > 0$, apply convexity of f with weights $\lambda a/c$, $\mu b/c$:

$$c f(s/c) \le \lambda a f(s/a) + \mu b f(s/b).$$

The remaining terms of (1) are linear in α, so each J_s is convex. The finite pointwise supremum in Lemma 1 is convex. This proves exactly the source's threshold-convexity assertion. □

Formal scope and research status

The Lean endpoint `SimonThreshold.source_threshold_convex` has only the hypothesis $\gamma > 1$. Its threshold is the literal real infimum over $\beta > 0$, with the universal real-power inequality on $0 < z < 1$. The logarithmic change of variables, positive perspective inequality, nonempty bounded supremum, positivity and exact infimum identification are proved. The proof uses pinned Lean 4.34.1 and mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612`. No optimizer-existence or differentiability premise remains in the endpoint.

This is a resolution of the separate final-paragraph question, not of MathDB 349140's main equivalence between complete monotonicity and the elementary criterion. Perspective convexity, convexity of suprema and exponential inequalities are standard tools; no new general convexity theorem is claimed. The accompanying proof files include independent review and reproducible verification instructions.

The actual arXiv v2 statement was read and visually inspected. The final publisher body remains inaccessible. The bounded literature screen includes the author's later publications, relevant gamma-ratio criteria and accessible citing bodies, with remaining access limits in the accompanying record. No worldwide priority claim is made.

References

[1] T. Simon, *Remark on a Mittag–Leffler function of Le Roy type*, arXiv:2012.11018v2 (2021), Section 3, p. 5. Published in Integral Transforms and Special Functions **33**(2) (2022), 108–114, doi:10.1080/10652469.2021.1913138.

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